



[STUDENT AI LAB · SAIL]

Student Workbook

Worksheet packets for all five modules — from $y = mx + b$ to a neural network you design, train, and explain yourself.

Modules 1–5 · Completed entirely in class — SAIL has no homework, on any module.

How This Workbook Works

This workbook holds the Student Worksheet Packet for every module of SAIL. Each packet is completed **during class**, **alongside the lesson** — your instructor will tell you when to turn to each worksheet. There is no homework on any module. Everything you practice, reason through, and plan happens in the room, on these pages.

THE SAIL RULE

Nothing gets calculated, dragged, or trained until your prediction is written. The prediction is part of the work — a prediction written after the fact is not a prediction.

EVERY WORKSHEET FOLLOWS THE SAME PATTERN

Part A — Guided practice, done with your instructor. **Part B — Independent practice**, done on your own, then checked with your table group. A short **reflection**, and an **extension** if you finish early.

The last worksheet in each packet is the **bridge worksheet** ("Looking Ahead") — it captures, in your own words and without notes, exactly what the next module builds on. Bring your packet back each session: it is your notes, your evidence, and where the next module starts.

Contents

MODULE	PACKET	YOU BUILD
1	Foundations of AI, Data, and Prediction	The two-part AI test, input/output maps, and $y = mx + b$ — on paper
2	The Linear Prediction Model	A slider-controlled line in Desmos, fit to real data with weight and bias
3	Neural Network Foundations	A complete forward pass through a real network — by hand, twice
4	Training a Neural Network	Loss by hand, then a live network trained on pictures you capture
5	Independent Build & Showcase	Your own experiment, your own network, and a 2-3 minute talk



MODULE 1 • STUDENT PACKET

Foundations of AI, Data, and Prediction

What makes an AI tick? The two-part definition, input and output, a first look at AI ethics — and the algebra everything else stands on.

Module 1 — Foundations of AI, Data, and Prediction

Student Worksheet Packet — complete alongside the lesson

Name: _____ Date: _____ Table Group: _____

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THE SAIL RULE

Nothing gets calculated until your prediction is written.

What is in this packet

#	WORKSHEET	WHEN YOU COMPLETE IT
1	AI or Not? — The Two-Step Sort	Segment 2 — during the lesson
2	Map the System — Input, Output, Data	Segment 4 — during the lesson
3	Bias, Data, and Consequences — Ethics Check	Segment 7 — during the lesson
4	Predict Before You Calculate — $y = mx + b$	Segment 8 — during the lesson
5	Looking Ahead to Module 2	Segment 8 — during the lesson
—	Module 1 Review (8 questions, no notes)	Segment 9 — end of class

AI or Not? — The Two-Step Sort

Complete during Segment 2, alongside the lesson.

DEFINITION REMINDER — AI

A computer system that improves at a task by learning patterns from data, rather than by following a fixed list of instructions written by a human.

THE TWO-STEP TEST

STEP	ASK YOURSELF	IF YES...
1	Did a human write the exact rule the system follows?	likely RULE-BASED
2	Did the system learn the pattern itself, from data?	likely AI

Part A — Guided Practice (do these two with your instructor)

#	SYSTEM DESCRIPTION	AI OR RULE-BASED?	WHICH PART OF THE DEFINITION TOLD YOU? (1 SENTENCE)
1	A vending machine that dispenses a snack when you insert the correct amount of money.		
2	A music app that learns which songs you skip and starts recommending different songs.		

Part B — Independent Practice (work alone, then check with your table group)

#	SYSTEM DESCRIPTION	AI OR RULE-BASED?	WHICH PART OF THE DEFINITION TOLD YOU? (1 SENTENCE)
3	A traffic light that changes color every 45 seconds, no matter what.		
4	A photo app that learns to recognize your face after you tag yourself a few times.		
5	A calculator that adds two numbers when you press “+”.		
6	An email inbox that learns from thousands of users marking messages as “spam” and filters similar messages automatically.		
7	A washing machine with a dial you turn to select “delicate” or “heavy duty.”		
8	A video platform that learns what kind of videos keep you watching longer and shows you more of them.		
9	An alarm clock that rings at the exact time you set it.		

#	SYSTEM DESCRIPTION	AI OR RULE-BASED?	WHICH PART OF THE DEFINITION TOLD YOU? (1 SENTENCE)
10	A customer-service chatbot that gets better at answering questions the more conversations it has.		

Reflection — answer in one sentence

Which of the ten was hardest to decide, and what made it hard?

Map the System — Input, Output, Data

Complete during Segment 4, alongside the lesson.

VOCABULARY REMINDER

Data: recorded information the system studied before it ever made a prediction. **Input:** the information given to the system right now, before it acts. **Output:** the result the system produces after processing the input.

Part A — Guided Practice (do System 1 with your instructor)

System 1 · A video game that adjusts its difficulty based on how well you're doing.

INPUT

OUTPUT

DATA IT LIKELY LEARNED FROM

Part B — Independent Practice (work alone 8 minutes, then check with your table group)

System 2 · A navigation app that suggests the fastest driving route.

INPUT

OUTPUT

DATA IT LIKELY LEARNED FROM

System 3 · A handwriting app that converts your handwritten notes into typed text.

INPUT

OUTPUT

DATA IT LIKELY LEARNED FROM

System 4 · A streaming service that shows you a "Recommended for You" row of shows.

INPUT

OUTPUT

DATA IT LIKELY LEARNED FROM

System 5 · A photo app that automatically sorts your pictures into albums like “Beach” or “Dogs.”

INPUT

OUTPUT

DATA IT LIKELY LEARNED FROM

Reflection

In one sentence, explain the difference between a system’s DATA and its INPUT. *They are not the same thing.*

Bias, Data, and Consequences — Ethics Check

Complete during Segment 7, alongside the lesson.

THE CORE IDEA

An AI system learns its patterns from data. If the data it learned from is incomplete, unbalanced, or unfair, the system will confidently reproduce that unfairness — and it will still *look* like it is working.

Part A — Guided Practice: Trace the Bias

Scenario (read with your instructor): A company builds an AI to screen job applications. It trains the system on the résumés of people the company has already hired — and almost everyone the company has ever hired came from three specific universities.

What DATA did this system learn from?

What pattern will it learn about “good” applicants?

Who gets harmed by this, and how?

Where did the bias actually come from — the code, or the data?

Part B — Independent Practice: Three More Systems

For each system, name one way the DATA could be biased, and one real consequence of that bias.

SYSTEM	HOW THE DATA COULD BE BIASED	ONE REAL CONSEQUENCE
A voice assistant trained mostly on recordings of adult male speakers.		
A medical AI trained on patient records from one wealthy hospital.		
A photo app trained mostly on pictures taken in bright daylight.		

Reflection

In two sentences: why is it dangerous that a biased AI system can still appear to be working perfectly?

Predict Before You Calculate

Complete during Segment 8, alongside the lesson.

THE FORMULA

$$y = mx + b$$

slope = m (the number multiplied by x) · **intercept = b** (the number added at the end)

Order of operations, every time: **identify** → **substitute** → **multiply** → **add**.

Part A — Guided Practice

Problem 1 · $y = 3x + 1$, find y when $x = 2$ (guided — with your instructor)

Identify

Slope (m) = _____ Intercept (b) = _____

My prediction (write BEFORE calculating)

Step 1 — Substitute

$y = (\quad)(\quad) + \quad$

Step 2 — Multiply

$y = \quad + \quad$

Step 3 — Add

$y = \quad$

Part B — Independent Practice

Problem 2 · $y = 3x + 1$, find y when $x = 5$

Identify

Slope (m) = _____ Intercept (b) = _____

My prediction (write BEFORE calculating)

Substitute → Multiply → Add

$y = (\quad)(\quad) + \quad = \quad + \quad = \quad$

Problem 3 · $y = -2x + 8$, find y when $x = 3$

Identify

Slope (m) = _____ Intercept (b) = _____

My prediction (write BEFORE calculating)

Substitute → Multiply → Add

$y = (\quad)(\quad) + \quad = \quad + \quad = \quad$

Problem 4 · $y = 0.5x + 4$, find y when $x = 10$

Identify

Slope (m) = _____ Intercept (b) = _____

My prediction (write BEFORE calculating)

Substitute → Multiply → Add

$y = (\quad)(\quad) + \quad = \quad + \quad = \quad$

Problem 5 • $y = -4x - 1$, find y when $x = 2$

Identify

Slope (m) = _____ Intercept (b) = _____

My prediction (write BEFORE calculating)

Substitute → Multiply → Add

$y = (\quad)(\quad) + \quad = \quad + \quad = \quad$

Reflection — Prediction vs. Reality

Circle one. Looking back, my predictions were generally: **A)** Higher than the actual answer **B)** Lower **C)** About right **D)** Mixed

What pattern do you notice about how the SLOPE affects the size of the output?

EXTENSION — IF YOU FINISH EARLY

Write your own equation in the form $y = mx + b$ with a **NEGATIVE** slope. Pick any x . Predict, then calculate. Then explain in one sentence what a negative slope does to the output as x gets bigger.

Looking Ahead to Module 2

Complete during Segment 8, alongside the lesson.

WHY THIS WORKSHEET EXISTS

Module 2 begins by assuming you already own four things from today. Write them down now, in your own words and without notes, so next session starts at full speed. This page is your bridge into Module 2 — bring it with you.

Part A — The Four Things I Own After Today

#	IN MY OWN WORDS...	MY ANSWER
1	What is Artificial Intelligence? (two-part definition)	
2	What is the difference between data, input, and output?	
3	In $y = mx + b$, what does m control? What does b control?	
4	Where does bias in an AI system usually come from?	

Part B — Next Session's Question

MY PREDICTION — WRITE THIS BEFORE YOU TEST

Next session you will drag a slider that changes m , and watch a line move on a screen in real time. Before you ever touch it: if you make m BIGGER, what do you predict happens to the line? If you make b bigger, what do you predict happens? Write both predictions now.

One thing I want to find out next session that I do not know yet:

Bring this packet to Module 2. Worksheet 5 is your notes, your evidence, and where Module 2 starts.

Module 1 Review

8 questions • 10 minutes • On your own. Notes closed.

Name: _____ Date: _____ Table Group: _____

Answer all 8 questions independently. Show your work for any calculation question.

Question 1 (Conceptual)

In your own words, write the two-part definition of Artificial Intelligence we used today.

Question 2 (Applied)

A spam filter looks at the words in an email and learns, from thousands of past examples, which patterns mean “spam.” Is this AI or rule-based? Justify your answer using the two-step test.

Question 3 (Conceptual)

For any AI system, define “input” and define “output” in your own words.

Question 4 (Applied)

A fitness app counts your steps using a fixed formula that never changes, no matter how many people use the app. Is this AI or rule-based? Justify your answer.

Question 5 (Conceptual)

Explain, in 1–2 sentences, why a lack of data would prevent an AI system from working well.

Question 6 (Conceptual)

Explain, in 1–2 sentences, where bias in an AI system usually comes from.

Question 7 (Calculation — show your work)

For the equation $y = 5x - 2$, calculate y when $x = 3$.

Question 8 (Calculation — show your work)

For the equation $y = -2x + 9$, calculate y when $x = 4$.



MODULE 2 · STUDENT PACKET

The Linear Prediction Model

Slope, intercept, and weighted sums — in Desmos, with your own hands.

Module 2 — The Linear Prediction Model

Slope, Intercept, and Weighted Sums in Desmos — Student Worksheet Packet

Name: _____ Date: _____ Table Group: _____

How to use this packet. Every worksheet is completed during class, alongside the segment it belongs to. There is no homework. Keep this packet open in front of you all day.

THE SAIL RULE

Nothing gets calculated and no slider gets dragged until your prediction is written.

What is in this packet

#	WORKSHEET	WHEN YOU COMPLETE IT
1	Rename the Parts — from m and b to w and b	Segment 2 — during the lesson
2	Plot and Judge — fitting a line by eye	Segment 4 — during the lesson
3	Build Your Model — Desmos sliders	Segment 5 — during the lesson
4	Linear or Nonlinear? — where a line breaks	Segment 7 — during the lesson
5	Looking Ahead to Module 3	Segment 8 — during the lesson
—	Module 2 Review (8 questions, no notes)	Segment 9 — end of class

Rename the Parts

Complete during Segment 2, alongside the lesson.

SAME MATH, NEW NAMES

Old algebra: $y = mx + b \rightarrow m = \text{slope}, b = \text{intercept}$

AI weighted sum: $y = wx + b \rightarrow w = \text{weight}, b = \text{bias}$

The calculation does not change. Only the vocabulary changes.

Part A — Guided Practice (do #1 with your instructor)

#	EQUATION / INPUT	OLD NAMES (M = , B =)	NEW NAMES (W = , BIAS =)	CALCULATION (MULTIPLY, THEN ADD)
1	$y = 2x + 5$ $x = 3$			$y = (\quad)(\quad) + \underline{\quad}$ $y = \underline{\quad} + \underline{\quad} = \underline{\quad}$

Part B — Independent Practice

#	EQUATION / INPUT	OLD NAMES (M = , B =)	NEW NAMES (W = , BIAS =)	CALCULATION (MULTIPLY, THEN ADD)
2	$y = -3x + 10$ $x = 2$			$y = (\quad)(\quad) + \underline{\quad} = \underline{\quad}$
3	$y = 0.5x + 6$ $x = 8$			$y = (\quad)(\quad) + \underline{\quad} = \underline{\quad}$
4	$y = 7x - 2$ $x = 1$			$y = (\quad)(\quad) + \underline{\quad} = \underline{\quad}$
5	$y = -1x + 4$ $x = 4$			$y = (\quad)(\quad) + \underline{\quad} = \underline{\quad}$

Reflection

In one sentence: how is the weighted-sum calculation DIFFERENT from $y = mx + b$? (*Careful — think before you answer.*)

Plot and Judge

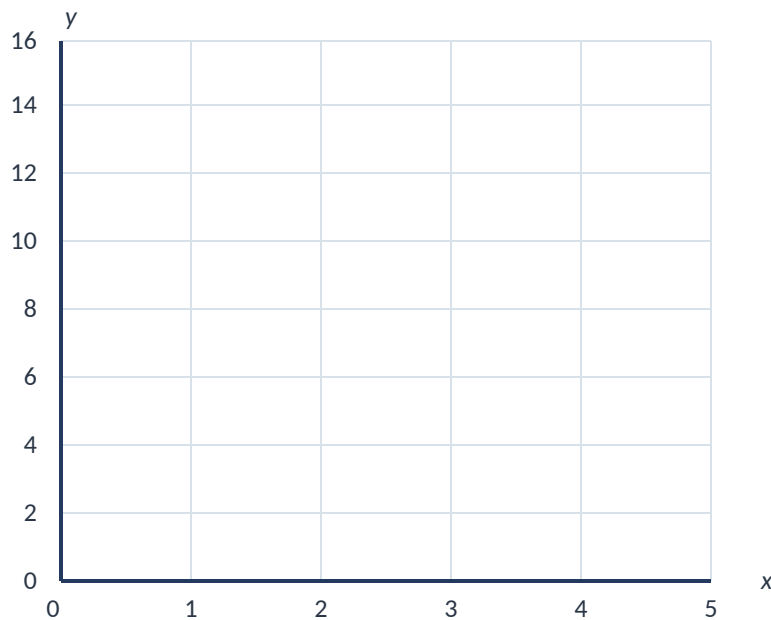
Complete during Segment 4, alongside the lesson.

DATA SET A

(1, 4) (2, 7) (3, 9) (4, 13) (5, 15) — Plot the points, then judge two candidate lines. Work alone for 8 minutes, then check with your table group.

Part A — Plot the Points

Plot each ordered pair as a dot. Label each dot with its coordinates.



Part B — Judge the Fit

For each candidate line, calculate the predicted y at each x , then judge the overall fit.

CANDIDATE LINE	$x=1$	$x=2$	$x=3$	$x=4$	$x=5$	GOOD / POOR FIT?	JUSTIFICATION (1 SENTENCE)
$y = 3x + 1$							
$y = 1x + 9$							

Reflection

Which line is better, and how did you decide — by looking, or by calculating? Which method do you trust more, and why?

Build Your Model (Desmos)

Complete during Segment 5, alongside the lesson.

DESMOS SETUP

1) Go to desmos.com/calculator. 2) Type $y = wx + b$ and press Enter. 3) Click “Add Slider” for both w and b . 4) You now have two sliders — w controls steepness, b controls starting height.

Part A — Predict, Then Test: The Weight Slider (w)

MY PREDICTION — WRITE THIS BEFORE YOU TEST

If I drag w to a LARGER POSITIVE number (like 5), the line will...

What actually happened — test it now, then write what you saw.

MY PREDICTION — WRITE THIS BEFORE YOU TEST

If I drag w into NEGATIVE numbers, the line will...

What actually happened — test it now, then write what you saw.

Part B — Predict, Then Test: The Bias Slider (b)

MY PREDICTION — WRITE THIS BEFORE YOU TEST

If I drag b to a larger positive number (like 8), the line will...

What actually happened — test it now, then write what you saw.

KEY QUESTION

Does changing b EVER change how steep the line is? Circle one: **YES / NO**

Explain your answer in one sentence:

Part C — Fit Your Line to Data Set A

Type these 5 points into a new row in Desmos: **(1,4), (2,7), (3,9), (4,13), (5,15)**. Then drag w and b until your line passes as close as possible to all 5 dots.

My final weight, w =

My final bias, b =

Class target: w near 3, b near 1 — how close are you?

Hand-check: using YOUR w and b , calculate y when $x = 3$ $y = (\text{ })(3) + \text{ } = \text{ } + \text{ } = \text{ }$

EXTENSION — IF YOU FINISH EARLY

Can you find a DIFFERENT pair of w and b values that fits Data Set A almost as well? Write them down. What does that tell you about whether there is only one “right” answer?

Linear or Nonlinear?

Complete during Segment 7, alongside the lesson.

HOW TO CHECK

Look at how much the output changes each time the input goes up by 1.

Gap stays the **SAME** size each time → **LINEAR**. Gap **GROWS** or **SHRINKS** → **NONLINEAR**.

Part A — Guided Practice (do #1 with your instructor)

#	DATA SET	GAP CALCULATIONS	LINEAR OR NONLINEAR?	FIT AFTER 60 SEC OF SLIDERS	JUSTIFICATION — CONNECT THE GAPS TO THE FIT
1	(1,5) (2,8) (3,11) (4,14)	8-5 = __ 11-8 = __ 14-11 = __			

Part B — Independent Practice

#	DATA SET	GAP CALCULATIONS	LINEAR OR NONLINEAR?	FIT AFTER 60 SEC OF SLIDERS	JUSTIFICATION — CONNECT THE GAPS TO THE FIT
2	(1,2) (2,6) (3,14) (4,30)	6-2 = __ 14-6 = __ 30-14 = __			
3	(1,10) (2,8) (3,6) (4,4)	8-10 = __ 6-8 = __ 4-6 = __			
4	(1,1) (2,8) (3,27) (4,64)	8-1 = __ 27-8 = __ 64-27 = __			

THE BIG IDEA — THE MOST IMPORTANT SENTENCE YOU WILL WRITE TODAY

In your own words: why can a single weighted sum ($y = wx + b$) NEVER fit a nonlinear data set, no matter how long you drag the sliders?

Looking Ahead to Module 3

Complete during Segment 8, alongside the lesson.

WHY THIS WORKSHEET EXISTS

Module 3 opens by assuming you already own the four statements below — it will not reteach them. Write them now, in your own words and without notes, so next session starts at full speed. Bring this page with you.

Part A — The Four Things I Own After Today

#	IN MY OWN WORDS...	MY ANSWER
1	What does the WEIGHT (w) control?	
2	What does the BIAS (b) control?	
3	Write the weighted-sum formula, and say what each letter means.	
4	What is the ONE thing a single weighted sum can never do?	

Part B — The Problem We Are Leaving Open on Purpose

Today you tried, and failed, to fit a straight line to curved data. That failure was not your fault — it was structural. Next session you will fix it.

MY PREDICTION — WRITE THIS BEFORE YOU TEST

Next session you will learn a tool that lets a model BEND instead of staying straight. Before you learn what it is: what do you think you would have to CHANGE about $y = wx + b$ to make it capable of bending? Guess freely — you are not expected to be right.

One thing I want to find out next session that I do not know yet:

Bring this packet to Module 3. Worksheet 5 is your notes, your evidence, and where Module 3 starts.

Module 2 Review

8 questions • 10 minutes • On your own. Notes closed.

Name: _____ Date: _____ Table Group: _____

Answer all 8 questions independently. Show your work for any calculation question.

Question 1 (Conceptual)

Explain how a weighted sum ($wx + b$) relates to last session's $y = mx + b$. Are they the same calculation, or different calculations?

Question 2 (Conceptual)

A model has weight w and bias b . Which one controls the STEEPNESS of the line, and which controls where the line starts when the input is 0?

Question 3 (Calculation — show your work)

For a model with weight = 6 and bias = -3, calculate the output when the input is 2.

Question 4 (Calculation — show your work)

For a model with weight = -2 and bias = 5, calculate the output when the input is 4.

Question 5 (Conceptual)

A data set's outputs increase by 5 every time the input increases by 1. Is this linear or nonlinear? Explain how you know.

Question 6 (Conceptual)

A data set has outputs 1, 3, 9, 27 as the input goes 1, 2, 3, 4. Linear or nonlinear? Explain using the size of the gaps.

Question 7 (Applied)

You drag your sliders for two full minutes and the line still doesn't get close to most of your points. Give TWO different possible explanations.

Question 8 (Conceptual)

In 1–2 sentences, explain why a single weighted sum can only ever produce a straight line, and why that becomes a limitation for some real-world data.



MODULE 3 · STUDENT PACKET

Neural Network Foundations

Neurons, layers, and the forward pass — the same math from Module 2, connected in a new way so it can finally bend.

Module 3 — Neural Network Foundations

Neurons, Layers, and the Forward Pass — Student Worksheet Packet

Name: _____ Date: _____ Table Group: _____

How to use this packet. Every worksheet is completed during class, alongside the segment it belongs to. There is no homework. Keep this packet open in front of you all day.

THE SAIL RULE

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What is in this packet

#	WORKSHEET	WHEN YOU COMPLETE IT
1	Label the Network — neurons and layers	Segment 2 — during the lesson
2	ReLU Practice — turning neurons on and off	Segment 4 — during the lesson
3	The Forward Pass — guided and independent	Segment 5 — during the lesson
4	Watch the Network Move — demo observation log	Segment 7 — during the lesson
5	Looking Ahead to Module 4	Segment 8 — during the lesson
—	Module 3 Review (8 questions, no notes)	Segment 9 — end of class

Label the Network

Complete during Segment 2, alongside the lesson.

Part A — Vocabulary in My Own Words

Do not copy from the board. Write each definition in words you would actually use.

TERM	MY DEFINITION, IN MY OWN WORDS
NEURON (what TWO things make it up?)	
LAYER (what makes a group of neurons a layer?)	
INPUT LAYER	
HIDDEN LAYER (why is it called “hidden”?)	
OUTPUT LAYER	

Part B — Guided Practice (do Network A with your instructor)

NETWORK	DESCRIPTION	HOW MANY LAYERS? NAME EACH ONE.	NEURONS IN EACH HIDDEN LAYER / OUTPUT LAYER
A	3 inputs connect to 1 neuron, which produces the final answer.		

Part C — Independent Practice (work with your partner — each of you fills out your own sheet)

NETWORK	DESCRIPTION	HOW MANY LAYERS? NAME EACH ONE.	NEURONS IN EACH HIDDEN LAYER / OUTPUT LAYER
B	2 inputs connect to 3 side-by-side neurons; those 3 neurons' outputs connect to 1 final neuron.		
C	2 inputs connect to 2 side-by-side neurons; those 2 outputs connect to 2 MORE side-by-side neurons; those outputs connect to 1 final neuron.		
D	4 inputs connect to 2 side-by-side neurons; those 2 outputs connect to 2 final, side-by-side output neurons.		

Reflection

A classmate says: "The input layer is just neurons like all the others." In one sentence, explain why that is not correct.

ReLU Practice

Complete during Segment 4, alongside the lesson.

THE RELU RULE — MEMORIZE THIS

If $z \geq 0 \rightarrow \text{ReLU}(z) = z$ (neuron is **ON** — the value passes through UNCHANGED)

If $z < 0 \rightarrow \text{ReLU}(z) = 0$ (neuron is **OFF** — the value is erased to zero)

There is no third case. ReLU never changes a positive number, and always erases a negative one.

Part A — Predict First, Then Calculate

Write your ON/OFF prediction BEFORE you write the output. Your prediction is part of the work.

#	WEIGHTED-SUM RESULT (z)	MY PREDICTION: ON OR OFF? (WRITE FIRST)	RELU OUTPUT
1	$z = 6$		
2	$z = -4$		
3	$z = -0.5$		
4	$z = 12$		
5	$z = -9$		
6	$z = 0$		
7	$z = 15$		
8	$z = -100$		

AFTER THE CLASS CHECK

Row 6 ($z = 0$) is the row most people get wrong. Which case does $z = 0$ fall into — ON or OFF? Explain in one sentence, using the exact rule.

Part B — Calculate the Weighted Sum, Then Apply ReLU

For each neuron: FIRST calculate $z = wx + b$ (show your multiply-then-add steps), THEN apply ReLU.

Neuron 1 · $w = 3$, $x = 2$, $b = -8$ (guided — with your instructor)

Step 1 — Calculate z	$z = (\quad)(\quad) + \quad = \quad + \quad = \quad$
Step 2 — My prediction	ON / OFF (circle one)
Step 3 — Apply ReLU	ReLU output = $\quad$

Neuron 2 • w = -2, x = 4, b = 10

Step 1 — Calculate z	$z = (\quad)(\quad) + \quad = \quad + \quad = \quad$
Step 2 — My prediction	ON / OFF (circle one)
Step 3 — Apply ReLU	ReLU output = $\quad$

Neuron 3 • w = 5, x = 0, b = -3

Step 1 — Calculate z	$z = (\quad)(\quad) + \quad = \quad + \quad = \quad$
Step 2 — My prediction	ON / OFF (circle one)
Step 3 — Apply ReLU	ReLU output = $\quad$

Neuron 4 • w = 1, x = 7, b = -6

Step 1 — Calculate z	$z = (\quad)(\quad) + \quad = \quad + \quad = \quad$
Step 2 — My prediction	ON / OFF (circle one)
Step 3 — Apply ReLU	ReLU output = $\quad$

Reflection

Neuron 3 has an input of $x = 0$. Explain in one sentence why its weight barely mattered — and what number DID decide whether it turned on.

The Forward Pass

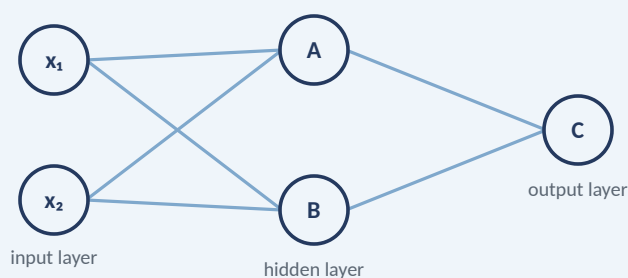
Complete during Segment 5, alongside the lesson.

FORWARD PASS

Calculating a network's full output by moving through it ONE LAYER AT A TIME, IN ORDER — where each layer's outputs become the next layer's inputs. This is not new math. It is a weighted sum, then ReLU, done several times in the right order.

Part A — Guided Forward Pass (with your instructor)

THE NETWORK



Structure: 2 inputs $\rightarrow$ 1 hidden layer with 2 neurons (A and B) $\rightarrow$ 1 output neuron (C, NO activation function).

Inputs: $x_1 = 4$, $x_2 = 1$ · **Neuron A:** weight on $x_1 = 2$, weight on $x_2 = -3$, bias = 1 · **Neuron B:** weight on $x_1 = -1$, weight on $x_2 = 2$, bias = -2 · **Neuron C:** weight on $a(A) = 1$, weight on $a(B) = 4$, bias = -2

STEP	WHAT YOU CALCULATE	YOUR WORK (SHOW MULTIPLY, THEN ADD)	RESULT
1	Neuron A weighted sum: $z(A) = (2)(x_1) + (-3)(x_2) + 1$		
2	Neuron A ReLU: ON or OFF? $\rightarrow$ output $a(A)$		
3	Neuron B weighted sum: $z(B) = (-1)(x_1) + (2)(x_2) + (-2)$		
4	Neuron B ReLU: ON or OFF? $\rightarrow$ output $a(B)$		
5	Output weighted sum (no ReLU): $z(C) = (1)(a(A)) + (4)(a(B)) + (-2)$		
6	FINAL NETWORK OUTPUT		

Part B — Independent Forward Pass (work alone, then check with your table group)

SAME STRUCTURE. NEW NUMBERS. NOBODY IS HELPING YOU THIS TIME.

Inputs: $x_1 = 3$, $x_2 = 2$ • **Neuron A:** weight on $x_1 = 1$, weight on $x_2 = -2$, bias = 4 • **Neuron B:** weight on $x_1 = 3$, weight on $x_2 = 1$, bias = -5 • **Neuron C:** weight on $a(A) = 2$, weight on $a(B) = 1$, bias = -3, no activation function

STEP	WHAT YOU CALCULATE	YOUR WORK (SHOW MULTIPLY, THEN ADD)	RESULT
1	$z(A) = (w_1)(x_1) + (w_2)(x_2) + \text{bias}$	$z(A) = (\quad)(\quad) + (\quad)(\quad) + \quad$	
2	ReLU on A. My prediction: ON / OFF	$a(A) = \text{ReLU}(z(A)) =$	
3	$z(B) = (w_1)(x_1) + (w_2)(x_2) + \text{bias}$	$z(B) = (\quad)(\quad) + (\quad)(\quad) + \quad$	
4	ReLU on B. My prediction: ON / OFF	$a(B) = \text{ReLU}(z(B)) =$	
5	$z(C) = (w)(a(A)) + (w)(a(B)) + \text{bias}$	$z(C) = (\quad)(\quad) + (\quad)(\quad) + \quad$	
6	FINAL NETWORK OUTPUT		

Part C — Reflection: Tracing a Zero Forward

In one of your two forward passes, a hidden neuron's ReLU turned it OFF (output = 0). When you multiplied that 0 by its weight in the output neuron's calculation, what did it contribute? Did the weight on that connection still matter? Explain in 2–3 sentences.

EXTENSION — IF YOU FINISH EARLY

Go back to Part B. Change Neuron A's bias from 4 to -10, and re-run only the steps that change. Does Neuron A flip from ON to OFF? What happens to the final output? Show your work.

Watch the Network Move

Complete during Segment 7, alongside the instructor demo.

YOU ARE WATCHING, NOT CLICKING

Your instructor will project a live neural network (mlu-explain.github.io/neural-networks). Your job is to connect what you SEE on the screen to the calculation you already did BY HAND on Worksheet 3.

Part A — Find It On the Screen

ON THE SCREEN I SEE...

IN MY HAND CALCULATION, THIS WAS...



The circles on the far LEFT

The circles in the MIDDLE

The circle(s) on the RIGHT

Each LINE connecting two circles

The number added inside each circle before
ReLU

Part B — The Big Question

In one or two sentences: why can THIS network fit shapes that last session's single weighted sum could not? Use the words **"activation function"** and **"straight line"** in your answer.

Part C — One Thing I Noticed

Write down one thing you saw on the projected screen that surprised you, or that you did not expect.

Looking Ahead to Module 4

Complete during Segment 8, alongside the lesson.

WHY THIS WORKSHEET EXISTS

Next session you take the controls of a real, running neural network. Module 4 assumes you already own everything below — it will not reteach it. Write it now, in your own words and without notes. Bring this page with you.

Part A — The Four Things I Own After Today

#	IN MY OWN WORDS...	MY ANSWER
1	A neuron is made of what two things?	
2	What does ReLU do to a NEGATIVE number? To a POSITIVE number?	
3	What is a hidden layer, and where does it sit?	
4	In a forward pass, what moves — and in what order?	

Part B — Thinking Questions

Answer in complete sentences. Reasoning matters more than being “right.”

1. A classmate says: “Adding more neurons and layers to a network ALWAYS makes it better.” Do you agree? Use today’s vocabulary (neuron, layer, activation function, weighted sum) to explain your thinking in 2–3 sentences.

2. Without an activation function, a network with 10 hidden neurons is no more powerful than ONE weighted sum. Explain why, in your own words. Use the phrase “**straight line**” somewhere in your answer.

3. Look back at Module 2's curved data set — the one whose gaps kept growing. Explain, in 2–3 sentences, how a network with ReLU could eventually fit that curve, even though a single weighted sum never could.

Part C — Next Session's Prediction

MY PREDICTION — WRITE THIS BEFORE YOU TEST

Next session you will be able to CHANGE the numbers inside a live network. Before you ever touch it: if you increased the bias on Neuron A in today's forward pass, which part of the calculation would change FIRST — and could that change flip the neuron from OFF to ON? Write your prediction now.

Bring this packet to Module 4. Worksheets 3 and 5 are your notes, your evidence, and where Module 4 starts.

Module 3 Review

8 questions · 10 minutes · On your own. Notes closed.

Name: _____ Date: _____ Table Group: _____

Answer all 8 questions independently. Show your work for any calculation question.

Question 1 (Conceptual)

In your own words, what TWO things is a neuron made of?

Question 2 (Conceptual)

State the ReLU rule completely. Your answer must cover negative inputs, positive inputs, and zero.

Question 3 (Calculation — show your work)

A neuron has $w = 4$, $x = 2$, and $b = -11$. Calculate z , then give the ReLU output.

Question 4 (Calculation — show your work)

A neuron has $w = -3$, $x = 5$, and $b = 20$. Calculate z , then give the ReLU output.

Question 5 (Conceptual)

A network has 2 inputs, then 4 side-by-side neurons, then 1 final neuron. How many layers does it have, and what is each one called?

Question 6 (Conceptual)

In a forward pass, can you calculate the OUTPUT neuron before you calculate the hidden neurons? Why or why not?

Question 7 (Applied)

In a forward pass, Hidden Neuron A's ReLU output is 0. Explain what that 0 contributes to the output neuron's calculation, and why.

Question 8 (Conceptual)

In 1-2 sentences, explain why chaining weighted sums together WITHOUT an activation function gives you nothing more powerful than a single weighted sum. Use the words "**straight line.**"



MODULE 4 · STUDENT PACKET

Training a Neural Network

Loss, learning, and live modification in the SAIL Architect Lab — where the weights on Module 3's worksheet actually came from.

Module 4 — Training a Neural Network

Loss, Learning, and Live Modification — Student Worksheet Packet

Name: _____ Date: _____ Table Group: _____

How to use this packet. Every worksheet is completed during class, alongside the segment it belongs to. There is no homework. Devices close during lecture segments.

THE SAIL RULE

Nothing gets trained until your prediction is written.

What is in this packet

#	WORKSHEET	WHEN YOU COMPLETE IT
1	Measuring Loss by Hand	Segment 2 — during the lesson
2	SAIL Architect Lab — Field Guide and Guided Run	Segments 4-5 — during the lesson
3	Modification Log — predict, modify, explain	Segment 7 — during the lesson
4	My Module 5 Project Plan	Segment 8 — during the lesson
—	Module 4 Review (8 questions, no notes)	Segment 9 — end of class

Measuring Loss by Hand

Complete during Segment 2, alongside the lesson. Devices closed.

THE THREE-STEP LOSS PROCEDURE

- 1 — **Predict:** plug x into the model to get a predicted output.
 - 2 — **Find the error:** error = prediction – correct answer. KEEP THE SIGN.
 - 3 — **Square and average:** square each error, add them up, divide by the number of points. That is the loss.
- LOWER loss = better model.** A loss of zero would mean every prediction was exactly right.

Part A — Guided Practice (with your instructor)

Data set: (1, 3) (2, 5) (3, 9) (4, 11)

Model A — prediction = $3x$

X	CORRECT ANSWER	PREDICTION	ERROR (PRED - CORRECT)	ERROR SQUARED
1				
2				
3				
4				

Loss of Model A = (____ + ____ + ____ + ____) $\div$ 4 = ____ $\div$ 4 = ____

Model B — prediction = $2x + 2$

X	CORRECT ANSWER	PREDICTION	ERROR (PRED - CORRECT)	ERROR SQUARED
1				
2				
3				
4				

Loss of Model B = (____ + ____ + ____ + ____) $\div$ 4 = ____ $\div$ 4 = ____

Circle the better model: MODEL A MODEL B

In one sentence, explain how you know — use the word “loss”:

Part B — Independent Practice (work alone, then check with your table group)

New data set: (1, 5) (2, 6) (3, 10) (4, 12)

Model C — prediction = $3x + 1$

X	CORRECT ANSWER	PREDICTION	ERROR (PRED - CORRECT)	ERROR SQUARED
1				
2				
3				
4				

Loss of Model C = (____ + ____ + ____ + ____) $\div$ 4 = _____

Model D — prediction = $2x + 3$

X	CORRECT ANSWER	PREDICTION	ERROR (PRED - CORRECT)	ERROR SQUARED
1				
2				
3				
4				

Loss of Model D = (____ + ____ + ____ + ____) $\div$ 4 = _____

Which model is better — circle one: MODEL C MODEL D NEITHER — THEY ARE TIED

CAREFUL

There may not be a winner. A full-credit answer states whatever the numbers actually show — even if that answer is “loss alone cannot separate these two models.” Do not force a winner.

Justify your choice in one sentence, using the word “loss”:

SAIL Architect Lab — Field Guide and Guided Run

Complete during Segments 4–5, alongside the lesson.

LAB URL

<https://cirqlar-official.github.io/SAIL-project-lab/> · Say YES to camera permission. **Do NOT refresh once class begins** — refreshing clears your captured pictures.

Starting Configuration Checklist

Match every setting below exactly. Check each box when confirmed.

SETTING	VALUE TO USE	✓
Class names	Rename to your chosen gesture pair — the whole class uses the same pair (default: Open Hand / Closed Fist)	
Hidden layers	2 (default)	
Activation Function	ReLU (Non-Linear) (default)	
Learning Rate	0.01 (default)	
Captured frames per class	At least 20–25 frames each. Move a little while recording — closer, farther, tilted.	

My gesture pair: Class 1 = _____ Class 2 = _____

BOX 1 — PREDICTION BEFORE THE FIRST TRAINING RUN · WRITE THIS BEFORE YOU TEST

When I press TRAIN MODEL, the Training Loss number will: **GO UP / GO DOWN / STAY THE SAME** (circle one). In one sentence, why?

Box 2 — Results of the First Training Run

Starting Training Loss (approximate)

Final Training Loss (approximate)

Was my Box 1 prediction correct? YES / NO

When I held each gesture up to the camera, what did the Live Inference Data bars do? (one sentence)

Box 3 — Reading My Training Loss Chart

Sketch the shape of your Training Loss chart in the box:

My chart matches — circle one:

CHART SHAPE	WHAT IT MEANS	WHAT I WOULD DO ABOUT IT
Falling STEEPLY	Still learning fast	Let it run
Falling slowly / flat and LOW	Finished learning	Halt — this is done
Flat and HIGH — never falls	Stuck — waiting will not help	Change something
Jumping up and down wildly	Learning rate too big	Lower the Learning Rate
Slowly, gently falling — still going down	Learning, but very slowly	Let it run, or raise the rate a little

In one sentence, what does YOUR chart tell you about YOUR network?

Box 4 — Guided Modification: Learning Rate → 0.001

Before any changes: do the **reset trick** (flip Activation Function to a different option, then back to ReLU). Then set Learning Rate to 0.001.

MY PREDICTION — WRITE THIS BEFORE YOU TEST

Compared to the first run, what will happen to how FAST the loss falls? How confident will the bars look after the same amount of training time?

Box 5 — Guided Modification Log Entry

WHAT I CHANGED	WHAT I PREDICTED	WHAT ACTUALLY HAPPENED	WHY IT HAPPENED
----------------	------------------	------------------------	-----------------

Modification Log

Complete during Segment 7, alongside the lesson.

STANDING PROCEDURE — EVERY SINGLE TIME

(1) Changing ONLY the Learning Rate? Do the reset trick first — flip the Activation Function away and back to ReLU. (2) Apply the change. (3) **WRITE YOUR PREDICTION.** (4) Press TRAIN MODEL. (5) Watch 30–45 sec. (6) HALT COMPILATION. (7) Hold up both gestures. (8) Fill in “What happened” and “Why.” **The “Why” column is the one that counts** — do not restate what happened; explain it.

Modification 1 — Change Activation Function to Linear

The change: Activation Function dropdown → Linear. (Network resets automatically — no reset trick needed.)

MY PREDICTION — WRITE THIS BEFORE YOU TEST

Will this network still be able to reliably tell your two gestures apart? What will the Training Loss chart do? Use what you know from Module 2.

WHAT I CHANGED

WHAT I PREDICTED

WHAT ACTUALLY HAPPENED

WHY IT HAPPENED

After logging: switch Activation back to ReLU and train once more — watch what changes immediately.

Modification 2 — Add Hidden Layers

The change: Reset trick (flip Activation away and back to ReLU), then use the layer-add button → 4 hidden layers.

MY PREDICTION — WRITE THIS BEFORE YOU TEST

More layers means more neurons, more weighted sums, more ReLUs. Will the loss get LOWER? Will it get there FASTER? Will the bars look more confident?

WHAT I CHANGED

WHAT I PREDICTED

WHAT ACTUALLY HAPPENED

WHY IT HAPPENED

Modification 3 — Remove Hidden Layers to One

The change: Use the layer-remove button → 1 hidden layer (the minimum).

MY PREDICTION — WRITE THIS BEFORE YOU TEST

One hidden layer, still with ReLU. Can ONE layer still tell your two gestures apart? Why or why not?

WHAT I CHANGED

WHAT I PREDICTED

WHAT ACTUALLY HAPPENED

WHY IT HAPPENED

Modification 4 — Break the Learning Rate

The change: Reset trick, restore hidden layers to 2, Activation to ReLU. Drag the Learning Rate slider to maximum: **0.100**.

MY PREDICTION — WRITE THIS BEFORE YOU TEST

This is TEN TIMES bigger than 0.01. Look at the U-curve on the board and picture that step size. What happens to the Training Loss chart?

WHAT I CHANGED

WHAT I PREDICTED

WHAT ACTUALLY HAPPENED

WHY IT HAPPENED

Reflection — Complete After All Four Modifications

1. You removed hidden layers (Mod 3), and you removed the activation function (Mod 1). Both made the network “simpler” — but they are FUNDAMENTALLY different limitations. In 2–3 sentences, explain the difference.

2. A classmate says: “So more layers is always better.” Using what you saw in Modification 2, explain in 2–3 sentences why that is not quite right.

Bring this worksheet to Module 5. It is your notes, your evidence, and where your Module 5 project starts. Your Module 5 presentation is one of these log entries, out loud.

My Module 5 Project Plan

Complete during Segment 8, alongside the lesson. This is your launchpad for next session.

WHAT MODULE 5 WILL ASK OF YOU

- Choose one gesture pair (or a new pair of categories) in the SAIL Architect Lab.
- Make at least THREE deliberate modifications, with a written prediction before each one.
- Get the network as confident as reasonably possible — and know WHY your final configuration works.
- Present for 2–3 minutes: what I tried, what I predicted, what happened, and why. **You have already done this. Four times. It is called Worksheet 3.**

Part A — My Plan

This is graded for COMPLETION, not correctness. Fill in every row. You will start Module 5 immediately — not spend the first thirty minutes deciding what to do.

PLANNING QUESTION

MY ANSWER

Which gesture pair (or new pair of categories) will I use?

What is my FIRST modification going to be — and what do I predict it will do?

What is my SECOND modification going to be — and why that one?

What is my THIRD modification going to be — and why that one?

What is ONE thing I want to find out next session that I do not know yet?

If my network gets STUCK (flat, high loss), what are the first TWO things I will try?

Part B — My Strongest Log Entry

Look back at Worksheet 3. Which of your four “Why” columns is your BEST explanation — the one where you explained a cause, not just an observation? Copy it here. This is the model for how you should sound next session.

Bring this packet to Module 5. Worksheet 3 (your log) and Worksheet 4 (your plan) are the two pages your entire Module 5 project is built on.

Module 4 Review

8 questions • 10 minutes • Devices closed. Notes closed.

Name: _____ Date: _____ Table Group: _____

Answer all 8 questions independently. Show your work for any calculation question.

Question 1 (Conceptual)

In your own words, what is loss? Your answer MUST say whether a higher or a lower loss is better.

Question 2 (Conceptual)

In order, name the four steps of the training loop.

1

2

3

4

Question 3 (Applied — show your work)

A model predicts 5, 8, and 10. The correct answers were 6, 8, and 12. Calculate the loss using the error-square-average method.

PREDICTION	CORRECT ANSWER	ERROR (PRED - CORRECT)	ERROR SQUARED
5	6		
8	8		
10	12		

Loss = (____ + ____ + ____) ÷ 3 = _____

Question 4 (Applied)

Model A has a loss of 0.9. Model B has a loss of 0.2. Which model is better, and how do you know?

Question 5 (Conceptual)

A network's Training Loss chart is flat and HIGH, and has been for a long time. Should the student wait longer, or change something? Explain in 1-2 sentences.

Question 6 (Conceptual)

You set the Activation Function to Linear and the network could not reliably tell your two gestures apart, no matter how long it trained. Explain WHY in 1-2 sentences. Use the word “**straight**” somewhere in your answer.

Question 7 (Conceptual)

Explain what the learning rate controls, and describe what goes wrong when it is set much too large.

Question 8 (Conceptual)

A classmate says: “My model is very confident on my own pictures, so it must be a good, fair model.” Explain in 2-3 sentences why that confidence alone does not guarantee this.



MODULE 5 · STUDENT PACKET

Independent Build & Showcase

Your network, your experiment, your explanation — today nobody tells you what to change.

Module 5 — Independent Build & Showcase

Designing, Running, and Presenting Your Own Network — Student Worksheet Packet

Name: _____ Date: _____ Table Group: _____

How to use this packet. Every worksheet is completed during class, alongside the segment it belongs to. There is no homework. Devices close during lecture segments. Today you run the lab yourself — the instructor is your studio coach, not your driver.

THE SAIL RULE

Nothing gets trained until your prediction is written.

What is in this packet

#	WORKSHEET	WHEN YOU COMPLETE IT
1	Design a Controlled Experiment	Segment 2 — during the lesson
2	Studio Log A — Build & Baseline	Segment 4 — during the lesson
3	Modification Log — change, predict, observe, explain	Segments 4 & 7 — during the lesson
4	Build Your Talk — the four-part frame	Segment 7 — during the lesson
5	Showcase — Peer Feedback Cards	Segment 8 — during the lesson
—	Module 5 Review (8 questions, no notes)	Segment 9 — end of class

Design a Controlled Experiment

Complete during Segment 2, alongside the lesson.

DEFINITION REMINDER — CONTROLLED EXPERIMENT

A test in which you change exactly one thing, hold everything else fixed, and write down what you expect to happen before you run it.

THE THREE MICRO-STEPS

- 1 — **Name the one variable** — the single setting you will change (activation function, number of hidden layers, or learning rate).
- 2 — **Fix the controls** — write down every other setting so it stays exactly the same.
- 3 — **Predict, then run** — write one sentence saying what you expect, then train.

My Experiment Plan

1. The one variable I will change:

2. The controls I will hold fixed (list them):

3. My prediction (before I train) — write one sentence starting with “*If my network...*”, and include a mechanism word: **straight-line limit, bending capacity, or step size**.

Studio Log A — Build & Baseline

Complete during Segment 4 (Studio Block A), at your device.

STUDIO PROCEDURE — KEEP THIS OPEN ALL BLOCK

1 — Capture your data. 2 — Write your prediction. 3 — Train and record. **Nothing gets trained until your prediction is written.**

Step 1 — Capture

What classes (gestures) did you capture? (two or more)

Class A: _____ Class B: _____ Class C (optional): _____

About how many pictures per class? _____

Step 2 — Baseline Settings and Prediction

SETTING	MY BASELINE VALUE
Activation function	
Number of hidden layers	
Learning rate	

MY PREDICTION FOR THE BASELINE RUN

What will the loss do?

Step 3 — Train and Record the Baseline

Did the Training Loss fall, stay flat, or bounce?

Roughly where did the loss settle (low / medium / high)?

Can the network tell my two classes apart? (yes / no / sometimes)

Modification Log

Complete during Studio Blocks A and B (Segments 4 & 7). One row per change.

THE DISCIPLINE

Change one thing → predict → observe → explain. Keep every other setting fixed so you know what caused the change. The “why” goes in terms of a mechanism — straight-line limit, bending capacity, or step size.

#	THE ONE THING I CHANGED	MY PREDICTION (BEFORE)	WHAT ACTUALLY HAPPENED	WHY (ONE SENTENCE)
1				
2				
3				
4				
5				

STUCK? FLAT, HIGH LOSS = CHANGE SOMETHING. DO NOT JUST WAIT.

Try these in order: **1)** Is your Activation Function on Linear? If your categories are shape-based, that's your problem — straight-line limit. Switch to ReLU. **2)** Already ReLU? Do the reset trick — flip the activation away and back — for a fresh random start, and train again. **3)** Still won't fall? Your data may be too similar between classes — recapture with more contrast.

Build Your Talk

Complete during Segment 7 (Studio Block B). This is your Presentation Builder.

THE FOUR-PART FRAME

A good technical talk says, in order: **(1)** the change you made, **(2)** what you predicted, **(3)** what actually happened, and **(4)** why — in plain language anyone could follow.

Draft Your 2–3 Minute Talk

Part 1 — The change. “The one thing I changed was...”

Part 2 — The prediction. “Before I ran it, I expected...”

Part 3 — What happened. “When I trained it, the loss...”

Part 4 — Why, in plain words. “The reason is...” (Define any term a newcomer wouldn’t know — for example, ReLU.)

JARGON CHECK

Circle any word in your talk that a person who never took this course wouldn’t know, and write a one-line plain definition to say right after it.

Showcase — Peer Feedback

Complete during Segment 8, as you listen to each talk in your pod.

THE THREE-POINT STANDARD (THE SAME ONE YOUR INSTRUCTOR SCORES)

For each talk, mark whether the speaker: **(1)** named the ONE thing they changed, **(2)** stated a real prediction, made before the run, with a reason, and **(3)** explained the WHY with a mechanism — straight-line limit, bending capacity, or step size — defining any course word a newcomer wouldn't know.

Speaker 1 — Name: _____

DID THE SPEAKER...	YES	NOT YET
Name the one thing they changed?		
State a prediction made before the run, with a reason?		
Explain the WHY with a mechanism word — not just “it got better”?		

One concrete thing — specific, not “good job” (e.g., “you nailed the why” or “I didn’t catch your prediction”):

Speaker 2 — Name: _____

DID THE SPEAKER...	YES	NOT YET
Name the one thing they changed?		
State a prediction made before the run, with a reason?		
Explain the WHY with a mechanism word — not just “it got better”?		

One concrete thing — specific, not “good job”:

Speaker 3 — Name: _____

DID THE SPEAKER...	YES	NOT YET
--------------------	-----	------------

Name the one thing they changed?

State a prediction made before the run, with a reason?

Explain the WHY with a mechanism word — not just “it got better”?

One concrete thing — specific, not “good job”:

Module 5 Review

8 questions · 8 minutes · Devices closed. Notes closed.

Name: _____ Date: _____ Table Group: _____

Answer all 8 questions independently. Show your work for the calculation question.

Question 1 (Conceptual)

What are the three rules of a controlled experiment in the SAIL Architect Lab? (One variable; everything else fixed; and...?)

Question 2 (Applied)

A classmate switches to Linear activation and raises the learning rate at the same time, then trains once. Why can't they conclude anything from this run? State the fix.

Question 3 (Applied / Calculation — show your work)

A tiny model predicts 5, 9, and 11. The correct answers were 4, 9, and 8. Calculate the loss using the error-square-average method.

Question 4 (Conceptual)

Your Training Loss chart has been flat and high for a long time. Name one specific thing you should change, and say why waiting longer will not help.

Question 5 (Conceptual)

Name the four parts of a good spoken technical explanation, in order.

Question 6 (Conceptual)

You are about to say the word “ReLU” to someone who never took this course. Write the short plain-language definition you would say right after it.

Question 7 (Conceptual)

A network is trained on ReLU activation with two hidden layers. You switch it to Linear and it can no longer tell your two hand shapes apart. Explain why, using the word “**straight.**”

Question 8 (Conceptual)

Your network reaches a very low loss and is extremely confident on your own captured pictures. Explain in two or three sentences why that alone does not make it a good, fair model for everyone.

THE ONE SENTENCE TO CARRY OUT OF THIS ROOM

A low loss does not mean a good model — it means a model that fits the data it was given. You didn't take anyone's word for that; you built the evidence yourself. The one question to always ask of any AI system: *what data was it given — and who wasn't in it?*